

INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

FST-L3

TD No3-Administration-systeme

Nom : EXUME

Prénom : Billy Rolph

Faculté : FST-INFORMATIQUE-L3

Prof : Ismael Saint Amour

Date : 22 MARS 2026

Introduction

Ce TD porte sur la supervision et la gestion des logs sous Linux sur un serveur Ubuntu. L'objectif est de surveiller les ressources système en temps réel, d'analyser les logs avec journalctl, de comprendre les logs applicatifs du serveur web nginx, de simuler un incident, et de configurer la rotation automatique des logs avec logrotate. L'ensemble des commandes a été exécuté sur une machine virtuelle Ubuntu Server accessible via SSH.

Résultats de l'exécution des commandes

1. Supervision en temps réel – commande top

```
billy@ubuntuserver: ~  
billy@ubuntuserver:~$ top  
top - 12:37:25 up 11:33, 2 users, load average: 0,00, 0,28, 0,82  
Tasks: 98 total, 1 running, 97 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0,0 us, 0,4 sy, 0,0 ni, 99,3 id, 0,0 wa, 0,0 hi, 0,4 si, 0,0 st  
MiB Mem : 1968,1 total, 426,9 free, 364,3 used, 1370,0 buff/cache  
MiB Swap: 2048,0 total, 2047,7 free, 0,3 used, 1603,8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3983	root	20	0	0	0	0	I	0,3	0,0	0:03.08	kworker/0:1-events
1	root	20	0	22632	13952	9680	S	0,0	0,7	0:15.46	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:00.04	kthreadd
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-rcu_p
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-slub_
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-netns
12	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-mm_pe
13	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_kthread
14	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_rude_kthread
15	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_trace_kthread
16	root	20	0	0	0	0	S	0,0	0,0	0:09.87	ksoftirqd/0
17	root	20	0	0	0	0	I	0,0	0,0	0:02.06	rcu_preempt
18	root	rt	0	0	0	0	S	0,0	0,0	0:00.64	migration/0
19	root	-51	0	0	0	0	S	0,0	0,0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kdevtmpfs
22	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	kworker/R-inet_
23	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kauditd
24	root	20	0	0	0	0	S	0,0	0,0	0:00.17	khungtaskd
26	root	20	0	0	0	0	S	0,0	0,0	0:00.00	oom_reaper

\

2. Supervision avec htop

```
billy@ubuntu:server: ~  
CPU[|||||] 0.8% Tasks: 24, 25 thr, 74 kthrs; 1 running  
Mem[|||||] 172M/1.92G Load average: 0.00 0.23 0.77  
Swp[|||||] 188K/2.00G Uptime: 11:34:12  
  
Main I/O  
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command  
12767 billy 20 0 8260 4672 3768 R 0.8 0.2 0:00.09 htop  
1 root 20 0 22632 13952 9680 S 0.0 0.7 0:15.46 /usr/lib/systemd/systemd --system --deseria  
354 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.67 /sbin/multipathd -d -s  
367 root 20 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/multipathd -d -s  
368 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/multipathd -d -s  
369 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/multipathd -d -s  
370 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/multipathd -d -s  
371 root RT 0 282M 27452 8760 S 0.0 1.4 0:04.53 /sbin/multipathd -d -s  
372 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/multipathd -d -s  
812 root 20 0 4236 2688 2432 S 0.0 0.1 0:00.95 /usr/sbin/cron -f -P  
813 messagebus 20 0 10316 6072 4672 S 0.0 0.3 0:01.94 @dbus-daemon --system --address=systemd: --  
825 root 20 0 18220 8972 7864 S 0.0 0.4 0:00.89 /usr/lib/systemd/systemd-logind  
854 root 20 0 104M 23092 13632 S 0.0 1.1 0:00.62 /usr/bin/python3 /usr/share/unattended-upgr  
927 root 20 0 104M 23092 13632 S 0.0 1.1 0:00.00 /usr/bin/python3 /usr/share/unattended-upgr  
1070 root 20 0 6948 4892 4076 S 0.0 0.2 0:00.04 /bin/login -p --  
1167 billy 20 0 20168 11288 9348 S 0.0 0.6 0:00.44 /usr/lib/systemd/systemd --user  
1168 billy 20 0 21152 3568 1844 S 0.0 0.2 0:00.00 (sd-pam)  
1179 billy 20 0 8656 5700 3976 S 0.0 0.3 0:00.19 -bash  
4964 root 20 0 14968 10688 8956 S 0.0 0.5 0:00.05 sshd: billy [priv]  
5061 billy 20 0 15128 7176 5176 S 0.0 0.4 0:00.46 sshd: billy@pts/1  
5064 billy 20 0 8648 5680 3968 S 0.0 0.3 0:00.07 -bash  
11445 root 19 -1 33936 12292 11224 S 0.0 0.6 0:00.10 /usr/lib/systemd/systemd-journald  
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

3. Vérification de l'espace disque – df -h

```
billy@ubuntu:server: ~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs           197M  1,1M  196M   1% /run  
/dev/mapper/ubuntu--vg-ubuntu--lv 12G   5,0G   5,7G  47% /  
tmpfs           985M    0  985M   0% /dev/shm  
tmpfs           5,0M    0   5,0M   0% /run/lock  
/dev/sda2       2,0G  200M   1,6G  11% /boot  
tmpfs           197M  12K  197M   1% /run/user/1000  
billy@ubuntu:server: ~$
```

4. Logs systemd – journalctl

```
billy@ubuntuuserver: ~$ journalctl
mars 09 02:44:04 ubuntuuserver kernel: Linux version 6.8.0-101-generic (buildd@lcy02-amd64-051) (x86_64-linux->
mars 09 02:44:04 ubuntuuserver kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-101-generic root=/dev/mapper/ub>
mars 09 02:44:04 ubuntuuserver kernel: KERNEL supported cpus:
mars 09 02:44:04 ubuntuuserver kernel: Intel GenuineIntel
mars 09 02:44:04 ubuntuuserver kernel: AMD AuthenticAMD
mars 09 02:44:04 ubuntuuserver kernel: Hygon HygonGenuine
mars 09 02:44:04 ubuntuuserver kernel: Centaur CentaurHauls
mars 09 02:44:04 ubuntuuserver kernel: zhaoxin Shanghai
mars 09 02:44:04 ubuntuuserver kernel: BIOS-provided physical RAM map:
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000000009fbff] usable
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000000009fc00-0x00000000000009ffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x0000000000000f0000-0x0000000000000fffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x000000000000100000-0x0000000000007ffeffff] usable
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x0000000007ffff0000-0x0000000007ffffffffff] ACPI data
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00ffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00ffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000ffff0000-0x00000000ffffffffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: NX (Execute Disable) protection: active
mars 09 02:44:04 ubuntuuserver kernel: APIC: Static calls initialized
mars 09 02:44:04 ubuntuuserver kernel: SMBIOS 2.5 present.
mars 09 02:44:04 ubuntuuserver kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
mars 09 02:44:04 ubuntuuserver kernel: Hypervisor detected: KVM
mars 09 02:44:04 ubuntuuserver kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
mars 09 02:44:04 ubuntuuserver kernel: kvm-clock: using sched offset of 2287911114354 cycles
mars 09 02:44:04 ubuntuuserver kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4d>
mars 09 02:44:04 ubuntuuserver kernel: tsc: Detected 2419.200 MHz processor
mars 09 02:44:04 ubuntuuserver kernel: e820: update [mem 0x00000000-0x000000ffff] usable ==> reserved
mars 09 02:44:04 ubuntuuserver kernel: e820: remove [mem 0x000a0000-0x0000ffff] usable
mars 09 02:44:04 ubuntuuserver kernel: last_pfn = 0x7ffff0 max_arch_pfn = 0x400000000
```

5. Logs système principaux – journalctl -xe

```
billy@ubuntuuserver: ~$ journalctl -xe
Defined-By: systemd
Support: http://www.ubuntu.com/support

The unit fwupd.service has successfully entered the 'dead' state.
mars 22 12:35:35 ubuntuuserver PackageKit[12148]: daemon quit
mars 22 12:35:35 ubuntuuserver systemd[1]: packagekit.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: http://www.ubuntu.com/support

The unit packagekit.service has successfully entered the 'dead' state.
mars 22 12:40:01 ubuntuuserver systemd[1]: Starting sysstat-collect.service - system activity accounting tool.
Subject: L'unité (unit) sysstat-collect.service a commencé à démarrer
Defined-By: systemd
Support: http://www.ubuntu.com/support

L'unité (unit) sysstat-collect.service a commencé à démarrer.
mars 22 12:40:01 ubuntuuserver systemd[1]: sysstat-collect.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: http://www.ubuntu.com/support

The unit sysstat-collect.service has successfully entered the 'dead' state.
mars 22 12:40:01 ubuntuuserver systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
Subject: L'unité (unit) sysstat-collect.service a terminé son démarrage
Defined-By: systemd
Support: http://www.ubuntu.com/support
```

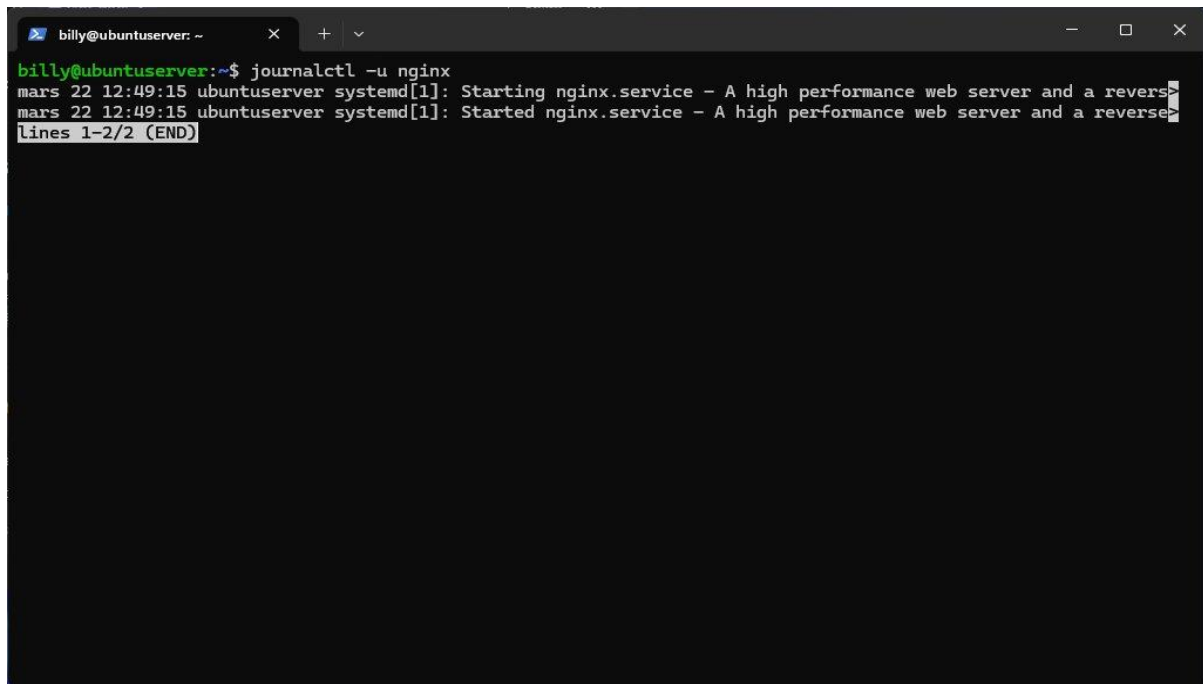
6. Logs du dernier démarrage – journalctl -b

```
billy@ubuntuuserver: ~$ journalctl -b
mars 09 02:44:04 ubuntuuserver kernel: Linux version 6.8.0-101-generic (buildd@lcy02-amd64-051) (x86_64-linux->
mars 09 02:44:04 ubuntuuserver kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-101-generic root=/dev/mapper/ub->
mars 09 02:44:04 ubuntuuserver kernel: KERNEL supported cpus:
mars 09 02:44:04 ubuntuuserver kernel: Intel GenuineIntel
mars 09 02:44:04 ubuntuuserver kernel: AMD AuthenticAMD
mars 09 02:44:04 ubuntuuserver kernel: Hygon HygonGenuine
mars 09 02:44:04 ubuntuuserver kernel: Centaur CentaurHauls
mars 09 02:44:04 ubuntuuserver kernel: zhaoxin Shanghai
mars 09 02:44:04 ubuntuuserver kernel: BIOS-provided physical RAM map:
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x0000000001000000-0x0000000007ffefff] usable
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x0000000007ffff0000-0x0000000007ffffffffff] ACPI data
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
mars 09 02:44:04 ubuntuuserver kernel: BIOS-e820: [mem 0x00000000ffff0000-0x00000000ffffffffff] reserved
mars 09 02:44:04 ubuntuuserver kernel: NX (Execute Disable) protection: active
mars 09 02:44:04 ubuntuuserver kernel: APIC: Static calls initialized
mars 09 02:44:04 ubuntuuserver kernel: SMBIOS 2.5 present.
mars 09 02:44:04 ubuntuuserver kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
mars 09 02:44:04 ubuntuuserver kernel: Hypervisor detected: KVM
mars 09 02:44:04 ubuntuuserver kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
mars 09 02:44:04 ubuntuuserver kernel: kvm-clock: using sched offset of 2287911114354 cycles
mars 09 02:44:04 ubuntuuserver kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4d->
mars 09 02:44:04 ubuntuuserver kernel: tsc: Detected 2419.200 MHz processor
mars 09 02:44:04 ubuntuuserver kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
mars 09 02:44:04 ubuntuuserver kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
```

7. Filtrage des erreurs – journalctl -p err

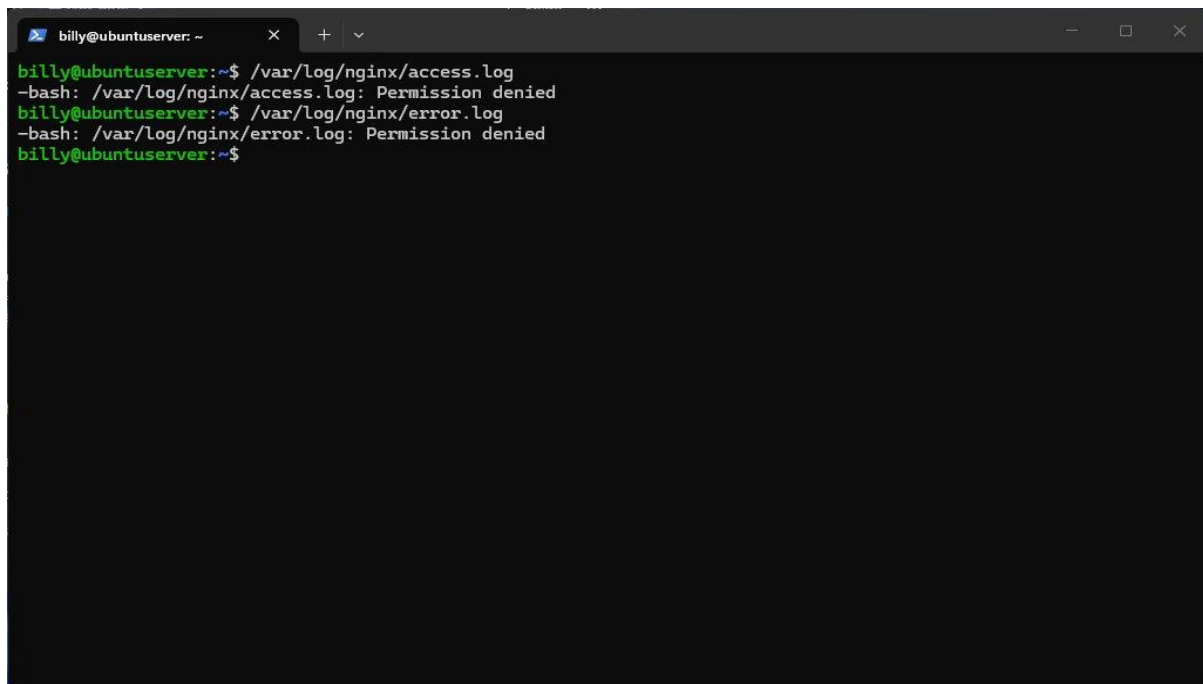
```
billy@ubuntuuserver: ~$ journalctl -p err
mars 09 02:44:04 ubuntuuserver kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
mars 09 02:44:08 ubuntuuserver kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an uns-
mars 09 02:44:08 ubuntuuserver kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
mars 09 02:44:08 ubuntuuserver kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphic-
mars 09 02:46:28 ubuntuuserver agetty[905]: tty1: invalid character 0x1b in login name
mars 09 02:47:00 ubuntuuserver agetty[1066]: tty1: invalid character 0x1b in login name
mars 09 02:47:24 ubuntuuserver login[1068]: PAM unable to dlopen(pam_lastlog.so): /usr/lib/security/pam_lastlo-
mars 09 02:47:24 ubuntuuserver login[1068]: PAM adding faulty module: pam_lastlog.so
mars 09 02:47:24 ubuntuuserver login[1070]: PAM unable to dlopen(pam_lastlog.so): /usr/lib/security/pam_lastlo-
mars 09 02:48:03 ubuntuuserver login[1070]: PAM adding faulty module: pam_lastlog.so
mars 16 21:11:14 ubuntuuserver kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 150s! [kworker/0:0:3933]
mars 16 21:42:26 ubuntuuserver kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 1740s! [dpkg-divert:3979]
mars 16 21:11:14 ubuntuuserver systemd[1]: systemd-journald.service: Watchdog timeout (limit 3min)!
lines 1-13/13 (END)
```

8. Logs du service nginx – journalctl -u nginx

A terminal window titled 'billy@ubuntu: ~' with standard window controls. The terminal shows the command 'journalctl -u nginx' being executed. The output displays two log entries from 'systemd[1]' regarding the 'nginx.service' on 'mars 22 12:49:15'. The first entry says 'Starting nginx.service - A high performance web server and a reverse proxy', and the second says 'Started nginx.service - A high performance web server and a reverse proxy'. The output is truncated with 'lines 1-2/2 (END)' at the bottom.

```
billy@ubuntu: ~$ journalctl -u nginx
mars 22 12:49:15 ubuntu systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy
mars 22 12:49:15 ubuntu systemd[1]: Started nginx.service - A high performance web server and a reverse proxy
lines 1-2/2 (END)
```

9. Accès aux logs nginx – Permission denied

A terminal window titled 'billy@ubuntu: ~' with standard window controls. The terminal shows two attempts to view nginx logs. The first attempt is to view '/var/log/nginx/access.log', which results in a 'Permission denied' error. The second attempt is to view '/var/log/nginx/error.log', which also results in a 'Permission denied' error. The prompt returns to the user after each error.

```
billy@ubuntu: ~$ /var/log/nginx/access.log
-bash: /var/log/nginx/access.log: Permission denied
billy@ubuntu: ~$ /var/log/nginx/error.log
-bash: /var/log/nginx/error.log: Permission denied
billy@ubuntu: ~$
```

10. Analyse des logs d'accès – tail et grep 404

```
billy@ubuntuuserver: ~  
billy@ubuntuuserver:~$ tail -f /var/log/nginx/access.log  
^C  
billy@ubuntuuserver:~$ grep 404 /var/log/nginx/access.log  
billy@ubuntuuserver:~$
```

11. Simulation d'incident – Arrêt de nginx

```
billy@ubuntuuserver: ~  
billy@ubuntuuserver:~$ sudo systemctl stop nginx  
billy@ubuntuuserver:~$ systemctl status nginx  
○ nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
   Active: inactive (dead) since Sun 2026-03-22 12:55:38 UTC; 11s ago  
     Duration: 6min 22.863s  
    Docs: man:nginx(8)  
   Process: 13048 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
   Process: 13049 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
   Process: 13192 ExecStop=/sbin/start-stop-daemon --quiet --stop --retry QUIT/5 --pidfile /run/nginx.pid (code=exited, status=0/SUCCESS)  
   Main PID: 13079 (code=exited, status=0/SUCCESS)  
    CPU: 58ms  
  
mars 22 12:49:15 ubuntuuserver systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server:  
mars 22 12:49:15 ubuntuuserver systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server:  
mars 22 12:55:38 ubuntuuserver systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server:  
mars 22 12:55:38 ubuntuuserver systemd[1]: nginx.service: Deactivated successfully.  
mars 22 12:55:38 ubuntuuserver systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server:  
Lines 1-16/16 (END)
```

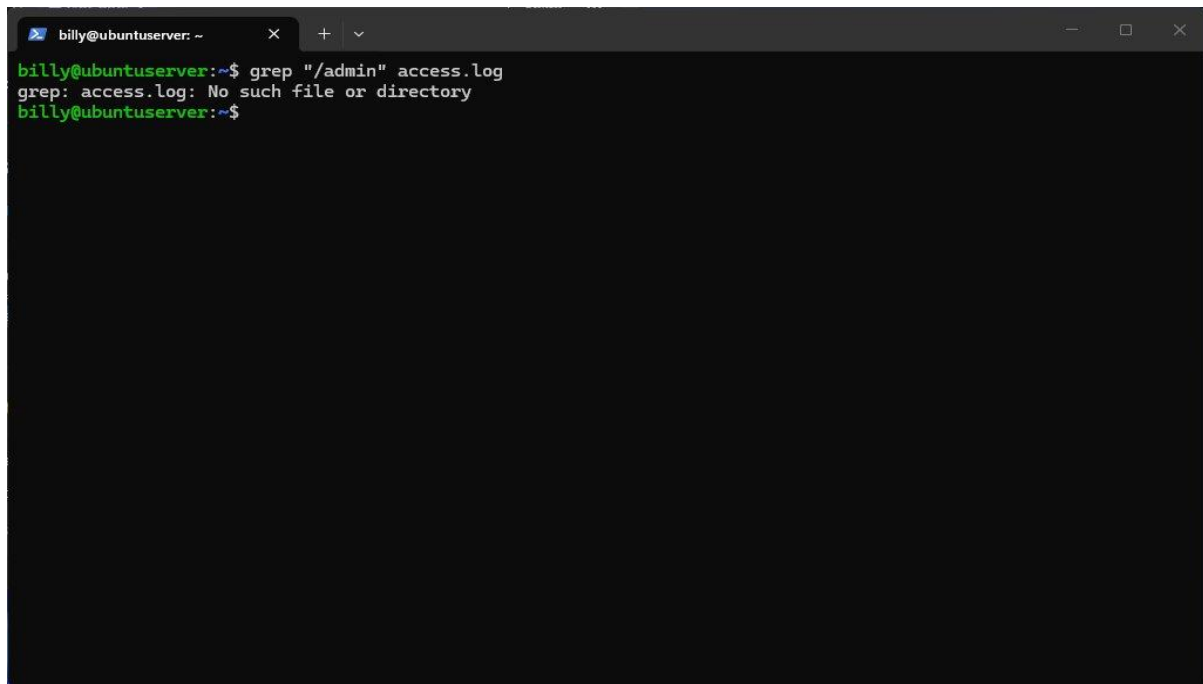
12. Observation des logs après l'incident – journalctl -u nginx -n 20

```
billy@ubuntuuserver: ~  
billy@ubuntuuserver:~$ journalctl -u nginx -n 20  
mars 22 12:49:15 ubuntuuserver systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server  
mars 22 12:49:15 ubuntuuserver systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server  
mars 22 12:55:38 ubuntuuserver systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server  
mars 22 12:55:38 ubuntuuserver systemd[1]: nginx.service: Deactivated successfully.  
mars 22 12:55:38 ubuntuuserver systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server  
lines 1-5/5 (END)
```

13. Redémarrage de nginx

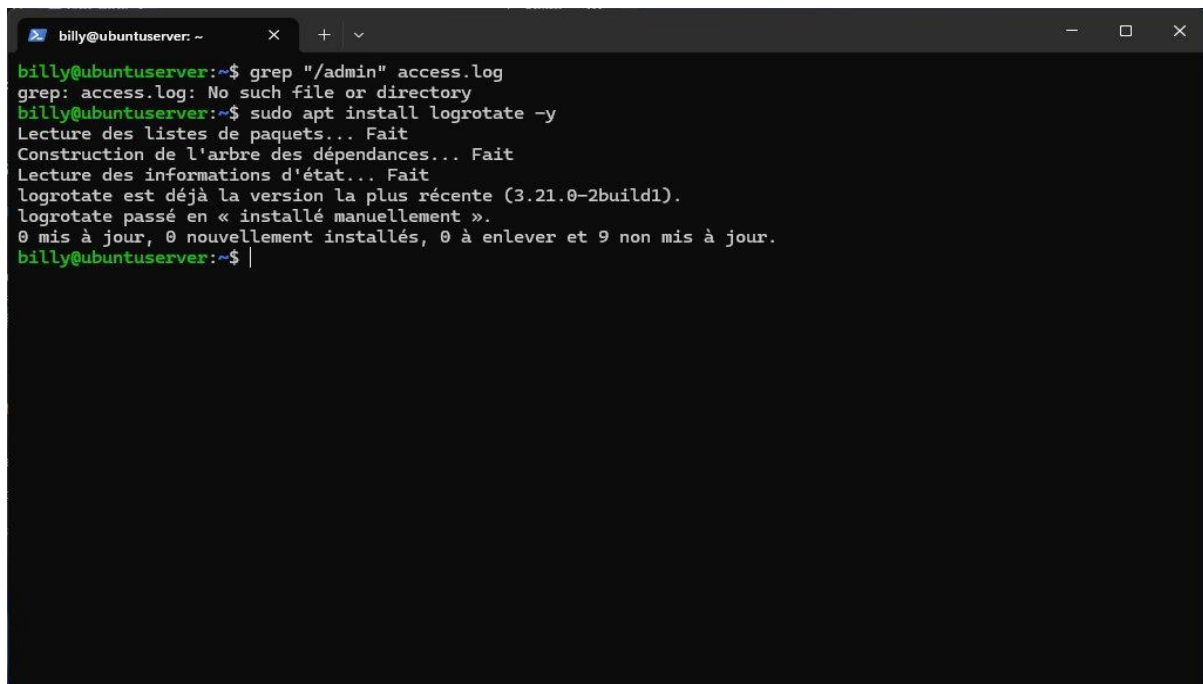
```
billy@ubuntuuserver: ~  
billy@ubuntuuserver:~$ sudo systemctl start nginx  
billy@ubuntuuserver:~$ systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
   Active: active (running) since Sun 2026-03-22 12:58:50 UTC; 10s ago  
     Docs: man:nginx(8)  
  Process: 13206 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
  Process: 13208 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
 Main PID: 13209 (nginx)  
    Tasks: 2 (limit: 2263)  
  Memory: 1.7M (peak: 1.9M)  
     CPU: 43ms  
   CGroup: /system.slice/nginx.service  
           └─13209 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"  
             └─13210 "nginx: worker process"  
  
mars 22 12:58:50 ubuntuuserver systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server  
mars 22 12:58:50 ubuntuuserver systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server  
lines 1-16/16 (END)
```

14. Analyse d'activité suspecte – grep "/admin"

A terminal window titled 'billy@ubuntu: ~' with standard window controls. The user enters the command 'grep "/admin" access.log'. The output is 'grep: access.log: No such file or directory'.

```
billy@ubuntu: ~$ grep "/admin" access.log
grep: access.log: No such file or directory
billy@ubuntu: ~$
```

15. Installation de logrotate

A terminal window titled 'billy@ubuntu: ~' with standard window controls. The user enters 'grep "/admin" access.log', which fails. Then they enter 'sudo apt install logrotate -y'. The output shows the package is already installed and up-to-date.

```
billy@ubuntu: ~$ grep "/admin" access.log
grep: access.log: No such file or directory
billy@ubuntu: ~$ sudo apt install logrotate -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
logrotate est déjà la version la plus récente (3.21.0-2build1).
logrotate passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 9 non mis à jour.
billy@ubuntu: ~$
```

16. Configuration globale de logrotate – /etc/logrotate.conf

```
billy@ubuntu:server: ~  
billy@ubuntu:server:~$ cat /etc/logrotate.conf  
# see "man logrotate" for details  
  
# global options do not affect preceding include directives  
  
# rotate log files weekly  
weekly  
  
# use the adm group by default, since this is the owning group  
# of /var/log/.  
su root adm  
  
# keep 4 weeks worth of backlogs  
rotate 4  
  
# create new (empty) log files after rotating old ones  
create  
  
# use date as a suffix of the rotated file  
#dateext  
  
# uncomment this if you want your log files compressed  
#compress  
  
# packages drop log rotation information into this directory  
include /etc/logrotate.d  
  
# system-specific logs may also be configured here.  
billy@ubuntu:server:~$ |
```

17. Liste des configurations logrotate – /etc/logrotate.d/

```
billy@ubuntu:server: ~  
billy@ubuntu:server:~$ ls /etc/logrotate.d/  
alternatives  apt      btmp      dpkg      rsyslog   ufw       wtmp  
apport        bootlog  cloud-init nginx      ubuntu-pro-client  unattended-upgrades
```

18. Création d'une rotation personnalisée – /etc/logrotate.d/devcloud

```

GNU nano 7.2 /etc/logrotate.d/devcloud *
/var/log/monitor.log {
    daily
    rotate 7
    compress
    missingok
    notifempty
    create 640 root adm
}

```

19. Test de la configuration logrotate – logrotate -d

[illegible]

20. Suite du test logrotate – Forcer la rotation avec logrotate -f

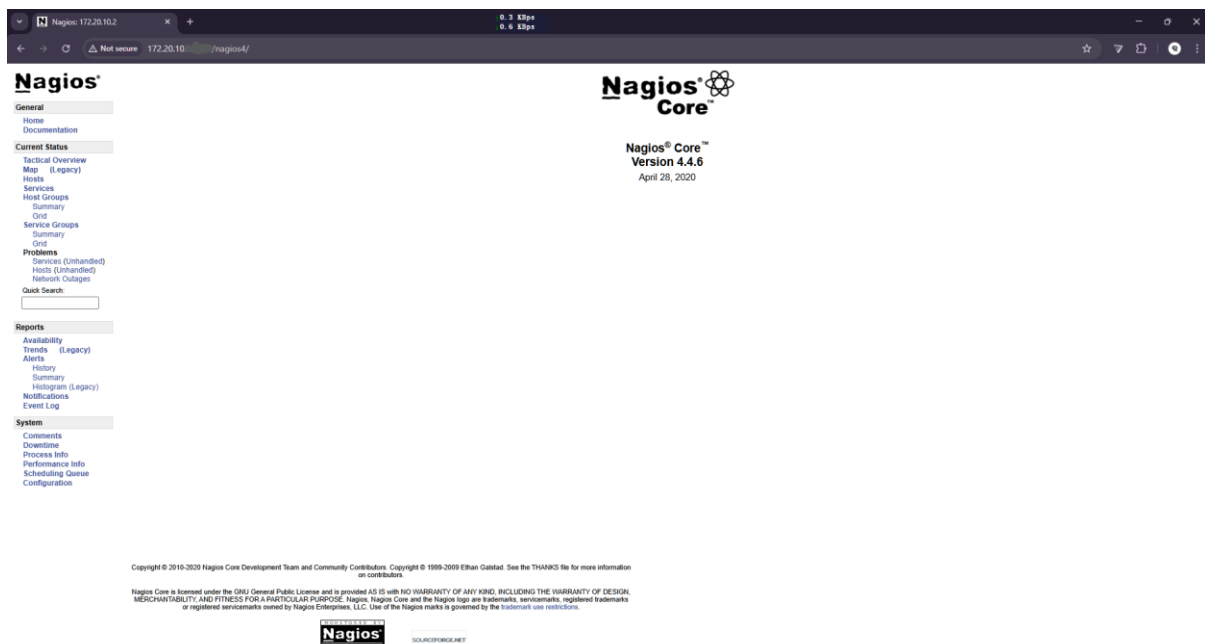
```
billy@ubuntuuser: ~  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Creating new state  
Handling 1 logs  
  
rotating pattern: /var/log/monitor.log after 1 days (7 rotations)  
empty log files are not rotated, old logs are removed  
considering log /var/log/monitor.log  
error: skipping "/var/log/monitor.log" because parent directory has insecure permissions (It's world writable  
or writable by group which is not "root") Set "su" directive in config file to tell logrotate which user/group  
should be used for rotation.  
Creating new state  
billy@ubuntuuser:~$ sudo logrotate -f /etc/logrotate.d/devcloud  
error: skipping "/var/log/monitor.log" because parent directory has insecure permissions (It's world writable  
or writable by group which is not "root") Set "su" directive in config file to tell logrotate which user/group  
should be used for rotation.  
billy@ubuntuuser:~$ |
```

21. Installation de Nagios

```
billy@ubuntuuser: ~  
billy@ubuntuuser:~$ sudo apt install nagios4 nagios-plugins -y  
Lecture des listes de paquets... Fait  
Construction de l'arbre des dépendances... Fait  
Lecture des informations d'état... Fait  
Note : sélection de « monitoring-plugins » au lieu de « nagios-plugins »  
Les paquets supplémentaires suivants seront installés :  
  apache2 apache2-bin apache2-data apache2-utils bsd-mailx javascript-common libapache2-mod-php  
  libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 libavahi-client3  
  libavahi-common-data libavahi-common3 libcups2t64 libdb1t64 libjs-jquery libldb2 liblockfile-bin  
  liblockfile1 liblua5.4-0 libmysqlclient21 libnet-snmp-perl libnsl2 libpq5 libradcli4 libsmbclient0  
  libsnmp-base libsnmp40t64 libtalloc2 libtdb1 libtevent0t64 liburiparser1 libwbclient0  
  monitoring-plugins-basic monitoring-plugins-common monitoring-plugins-standard mysql-common nagios-images  
  nagios4-cgi nagios4-common nagios4-core php-common php8.3-cli php8.3-common php8.3-opcache php8.3-readline  
  postfix python3-gpg python3-ldb python3-markdown python3-samba python3-talloc python3-tdb rpcbind  
  samba-common samba-common-bin samba-dsdb-modules samba-libs smbclient snmp ssl-cert  
Paquets suggérés :  
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser php-pear cups-common  
  libcrypt-des-perl libdigest-hmac-perl libio-socket-inet6-perl snmp-mibs-downloader icinga2  
  nagios-plugins-contrib fping qstat nagios-nrpe-plugin postfix-cdb postfix-doc postfix-ldap postfix-lmdb  
  postfix-mta-sts-resolver postfix-mysql postfix-pcre postfix-pgsql postfix-sqlite procmail sasl2-bin  
  | dovecot-common python-markdown-doc heimdal-clients python3-dnspython cifs-utils  
Les NOUVEAUX paquets suivants seront installés :  
  apache2 apache2-bin apache2-data apache2-utils bsd-mailx javascript-common libapache2-mod-php  
  libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 libavahi-client3  
  libavahi-common-data libavahi-common3 libcups2t64 libdb1t64 libjs-jquery libldb2 liblockfile-bin  
  liblockfile1 liblua5.4-0 libmysqlclient21 libnet-snmp-perl libnsl2 libpq5 libradcli4 libsmbclient0  
  libsnmp-base libsnmp40t64 libtalloc2 libtdb1 libtevent0t64 liburiparser1 libwbclient0 monitoring-plugins  
  monitoring-plugins-basic monitoring-plugins-common monitoring-plugins-standard mysql-common nagios-images  
  nagios4 nagios4-cgi nagios4-common nagios4-core php-common php8.3-cli php8.3-common php8.3-opcache  
  php8.3-readline postfix python3-gpg python3-ldb python3-markdown python3-samba python3-talloc python3-tdb
```

```
billy@ubuntu: ~  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
billy@ubuntu:~$ sudo systemctl status nagios4  
● nagios4.service - nagios4  
   Loaded: loaded (/usr/lib/systemd/system/nagios4.service; disabled; preset: enabled)  
   Active: active (running) since Sun 2026-03-22 13:11:12 UTC; 4min 47s ago  
     Docs: man:nagios4  
    Main PID: 23890 (nagios4)  
       Tasks: 6 (limit: 2263)  
      Memory: 2.2M (peak: 3.1M)  
         CPU: 224ms  
    CGroup: /system.slice/nagios4.service  
            └─23890 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg  
              └─23891 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
                └─23892 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
                  └─23893 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
                    └─23894 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh  
                      └─23895 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg  
  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23894;pid=23894  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23894;pid=23894  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23893;pid=23893  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23891;pid=23891  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23893;pid=23893  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23892;pid=23892  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23891;pid=23891  
mars 22 13:11:12 ubuntu: nagios4[23890]: wproc: Registry request: name=Core Worker 23892;pid=23892  
mars 22 13:11:13 ubuntu: nagios4[23890]: Successfully launched command file worker with pid 23895  
mars 22 13:11:13 ubuntu: nagios4[23890]: Successfully launched command file worker with pid 23895  
billy@ubuntu:~$
```

Interface web Nagios



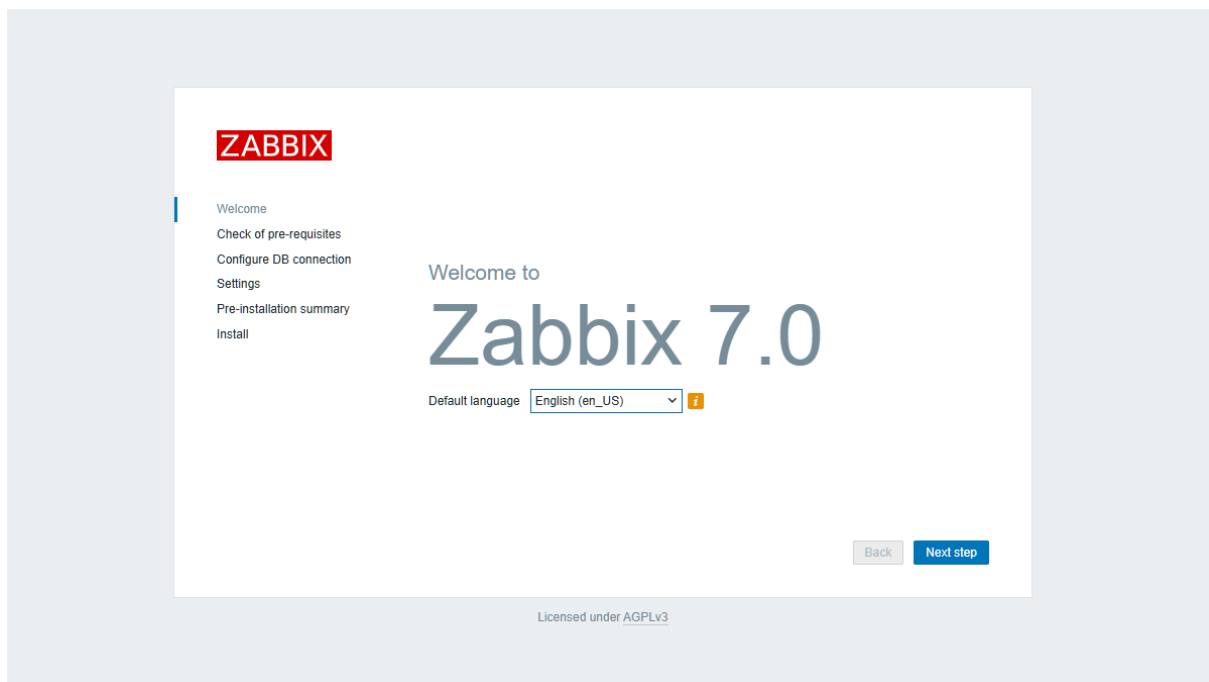
Téléchargement et installation du dépôt Zabbix

```
billy@ubuntu-server: ~  
Paramétrage de libfcgi-perl (0.82+ds-3build2) ...  
Paramétrage de php8.3-xml (8.3.6-0ubuntu0.24.04.7) ...  
  
Creating config file /etc/php/8.3/mods-available/dom.ini with new version  
  
Creating config file /etc/php/8.3/mods-available/simplexml.ini with new version  
  
Creating config file /etc/php/8.3/mods-available/xml.ini with new version  
  
Creating config file /etc/php/8.3/mods-available/xmlreader.ini with new version  
  
Creating config file /etc/php/8.3/mods-available/xmlwriter.ini with new version  
  
Creating config file /etc/php/8.3/mods-available/xsl.ini with new version  
Paramétrage de zabbix-agent (1:7.0.24-1ubuntu24.04) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/zabbix-agent.service → /usr/lib/systemd/system/zabbix-agent.service.  
Paramétrage de fping (5.1-1) ...  
Paramétrage de liburi-perl (5.27-1) ...  
Paramétrage de php8.3-gd (8.3.6-0ubuntu0.24.04.7) ...  
  
Creating config file /etc/php/8.3/mods-available/gd.ini with new version  
Paramétrage de php-xml (2:8.3+93ubuntu2) ...  
Paramétrage de mysql-server-8.0 (8.0.45-0ubuntu0.24.04.1) ...  
update-alternatives: utilisation de « /etc/mysql/mysql.cnf » pour fournir « /etc/mysql/my.cnf » (my.cnf) en mode automatique  
Renaming removed key_buffer and myisam-recover options (if present)  
mysqld will log errors to /var/log/mysql/error.log  
  
Progression : [ 81%] [#####.....]
```

```
billy@ubuntu-server: ~  
billy@ubuntu-server:~$ wget https://repo.zabbix.com/zabbix/6.4/ubuntu/pool/main/z/zabbix-release/zabbix-release_6.4-1+ubuntu22.04_all.deb  
--2026-03-22 13:42:34-- https://repo.zabbix.com/zabbix/6.4/ubuntu/pool/main/z/zabbix-release/zabbix-release_6.4-1+ubuntu22.04_all.deb  
Resolving repo.zabbix.com (repo.zabbix.com)... 178.128.6.101, 2604:a880:2:d0::2062:d001  
Connecting to repo.zabbix.com (repo.zabbix.com)|178.128.6.101|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 3744 (3.7K) [application/octet-stream]  
Saving to: 'zabbix-release_6.4-1+ubuntu22.04_all.deb'  
  
zabbix-release_6.4-1+ubuntu 100%[=====] 3,66K --.-KB/s in 0s  
  
2026-03-22 13:42:35 (201 MB/s) - 'zabbix-release_6.4-1+ubuntu22.04_all.deb' saved [3744/3744]  
  
billy@ubuntu-server:~$ sudo dpkg -i zabbix-release_6.4-1+ubuntu22.04_all.deb  
Sélection du paquet zabbix-release précédemment désélectionné.  
(Lecture de la base de données... 131050 fichiers et répertoires déjà installés.)  
Préparation du dépaquetage de zabbix-release_6.4-1+ubuntu22.04_all.deb ...  
Dépaquetage de zabbix-release (1:6.4-1+ubuntu22.04) ...  
Paramétrage de zabbix-release (1:6.4-1+ubuntu22.04) ...  
billy@ubuntu-server:~$ sudo apt update  
Atteint :1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Réception de :2 https://repo.zabbix.com/zabbix/6.4/ubuntu jammy InRelease [2 883 B]  
Atteint :3 http://ht.archive.ubuntu.com/ubuntu noble InRelease  
Atteint :4 http://ht.archive.ubuntu.com/ubuntu noble-updates InRelease  
Atteint :5 http://ht.archive.ubuntu.com/ubuntu noble-backports InRelease  
Réception de :6 https://repo.zabbix.com/zabbix/6.4/ubuntu jammy/main Sources [23,8 kB]  
Réception de :7 https://repo.zabbix.com/zabbix/6.4/ubuntu jammy/main amd64 Packages [63,5 kB]  
Réception de :8 https://repo.zabbix.com/zabbix/6.4/ubuntu jammy/main all Packages [12,7 kB]  
103 ko réceptionnés en 2s (51,6 ko/s)
```

```
billy@ubuntu:~$ sudo systemctl enable zabbix-server
Synchronizing state of zabbix-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable zabbix-server
Created symlink /etc/systemd/system/multi-user.target.wants/zabbix-server.service → /usr/lib/systemd/system/zabbix-server.service.
billy@ubuntu:~$ sudo systemctl start zabbix-server
Command 'sudo' not found, but can be installed with:
sudo apt install sudo
billy@ubuntu:~$ sudo systemctl start zabbix-server
billy@ubuntu:~$
```

Interface web Zabbix



Questions

1. Quelle est la différence entre logs système et logs applicatifs ?

Les logs système enregistrent les événements du noyau et des services de l'OS (gérés par systemd/journalctl). Les logs applicatifs sont produits par des programmes spécifiques comme nginx, et stockés dans des fichiers dédiés sous /var/log/.

2. Pourquoi centraliser les logs en entreprise ?

La centralisation permet de consulter tous les logs depuis un seul endroit, de détecter plus facilement les incidents, de corréler des événements sur plusieurs machines, et de faciliter les audits de sécurité.

3. Que se passe-t-il si les logs deviennent trop volumineux ?

Des logs trop volumineux peuvent saturer le disque, ralentir le système et rendre les fichiers difficiles à lire. C'est pour cela que logrotate existe : il archive, compresse et supprime les anciens logs automatiquement.

4. Quel est le rôle de logrotate ?

Logrotate automatise la gestion des fichiers de logs : il les fait tourner à intervalles réguliers (quotidien, hebdomadaire), les compresse, en conserve un nombre défini et crée de nouveaux fichiers vides pour la suite.

5. Pourquoi surveiller la RAM et le CPU ?

Surveiller la RAM et le CPU permet de détecter une surcharge ou un processus anormal avant que le système ne devienne instable ou inaccessible. C'est essentiel pour garantir la disponibilité des services.

Conclusion

Ce TD a permis de mettre en pratique les principaux outils de supervision et de gestion des logs sous Linux. Les commandes top, htop, journalctl et logrotate constituent des outils indispensables pour tout administrateur système. La simulation d'incident avec nginx a illustré concrètement comment détecter, analyser et résoudre un problème de service.